Question

The incorrect statement about electrochemistry is

A. The study of generation of electricity by spontaneous chemical reactions

B. The study of use of electrical energy to bring out a non-spontaneous chemical reaction

C. Reaction based on it is not so efficient and ecofriendly

D. Transmission of sensory signals through cells to brain are of electrochemical origin

Question

In the galvanic cell Cu | Cu^{2+} (1M) || Ag^{+} (1M) Ag the electrons will travel in the external circuit:

(A) from Ag to Cu

(B) from Cu to Ag

(C) electrons do not travel in the external circuit

(D) in any direction

Question

$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$
The above redox reaction is used in

(A) Galvanic cell

(B) Daniell cell

(C) Voltaic cell

(D) All of these

Question

In general, in a Galvanic cell

A. Electrons flow from positive to negative electrode

B. Current flow from positive to negative electrode

C. Electrons and current flow in the same direction

D. All of the above are correct

Question

When applied external opposite potential in case of Daniell cell is less than 1.1 V, then

A. The cell behaves like an electrolytic cell

B. Electrons flow from zinc rod to copper rod

C. Zinc deposits at the zinc electrode and copper dissolves from copper rod

D. Current flow from zinc rod to copper rod

Question

In the construction of a salt bridge, saturated solution of KNO_3 is used because

(A) Velocity of K^+ & NO_3^- are same.

(B) Velocity of NO_3^- is greater than that of K^+.

(C) Velocity of K^+ is greater than that of NO_3^-.

(D) KNO_3 is highly soluble in water.

Question

What happens when applied external opposite potential in a Daniell cell reaches to 1.1 V?

A. Chemical reaction stops

B. Current starts flowing in a direction opposite to that of the electrons

C. Current starts flowing in the same direction as of electrons

D. The cell behaves like an electrolytic cell

Question

If a salt bridge is removed between the half cells, the voltage:

A. drops to zero

B. does not charge

C. increase gradually

D. increases rapidly

Question

Assertion (A): The electrochemical cell stops working after some time.

Reason (R): Electrode potentials of both electrodes become equal.

A. Both Assertion (A) and Reason (R) are true and the Reason (R) is the correct explanation of the Assertion (A).

B. Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of the Assertion (A).

C. Assertion (A) is true and Reason (R) is false.

D. Assertion (A) is false and Reason (R) is true.

Question

Which one of the following is not a function of a salt bridge?

A. To allow the flow of cations from one solution to the other.

B. To allow the flow of anions from one solution to the other.

C. To allow the electrons to flow from one solution to the other.

D. To maintain electrical neutrality of the two solutions.

Question

Statement-I: Salt bridge is used generally in the electrochemical cells.

Statement-II: The ions of the electrolyte used in the salt bridge should have nearly same transport numbers.

A. Statement-I and Statement-II both are correct.

B. Statement-I is correct but Statement-II is incorrect.

C. Statement-I is incorrect but Statement-II is correct.

D. Statement-I and Statement-II both are incorrect.

Question

If in a salt bridge KNO_3 solution is filled then.

A. from salt bridge NO_3^-(aq) ion goes to anode half cell and neutralizes the left Zn^{2+} formed.

B. from salt bridge NO_3^-(aq) ions goes in direction of anode half cell but doesn't go to the half cell of anode solution.

C. from salt bridge, NO_3^- ions goes in anode half cell solution and K^+ ions goes in cathode half cell solution.

D. in salt bridge, concentration of KNO_3 decreases.

Question

During the working of the Daniell cell, which of the following happens?

A. The size of the Zn rod as well as the intensity of the colour of $CuSO_4$ solution remain unchanged.

B. The size of the Zn rod is reduced and the blue colour of $CuSO_4$ solution becomes faint.

C. The size of the Zn rod remains same but the blue colour of $CuSO_4$ solution becomes faint.

D. The size of the Zn rod is reduced but there is no change in the intensity of colour of the $CuSO_4$ solution.

Question

Reaction: $Mg(s) + Co^{2+}(aq) \rightleftharpoons Mg^{2+}(aq) + Co(s)$ show the cell representation of the reaction:

A. $Co(s) \mid Co^{2+}(1M) \parallel Mg^{2+}(s) \mid Mg^{2+}(s)$

B. $Co(s) \mid Co^{2+}(1M) \parallel Mg(s) \mid Mg(aq)(1M)$

C. $Mg(s) \mid Mg^{2+}(1M) \parallel Co(aq) \mid Co^{2+}(1M)$

D. $Mg(s) \mid Mg^{2+}(1M) \parallel Co^{2+}(1M) \parallel Co(s)$

Question

Select the correct cell reaction of the cell $Pt(s) \mid Cl^-(aq) \mid Cl_2(g) \parallel Ag^+(aq) \mid Ag(s)$.

A. $Cl_2(g) + Ag^+(aq) \rightarrow Ag(s) + 2Cl^-(aq)$

B. $Cl_2(g) + Ag(s) \rightarrow 2Cl^-(aq) + Ag^+(aq)$

C. $2Cl^-(aq) + 2Ag^+(aq) \rightarrow 2Ag(s) + Cl_2(g)$

D. $AgCl(s) \rightarrow Ag^+(aq) + Cl^-(aq)$

Question

At 25°C temperature, which of the following reaction is correct for electrochemical cell?

Cell: Pt | Br_2(l) | Br^-(aq) || Cl^-(aq) | Cl_2(g) | Pt

A. $2Br^-(aq) + Cl_2(g) \rightarrow 2Cl^-(aq) + Br_2(l)$

B. $Br_2(l) + 2Cl^-(aq) \rightarrow Cl_2(g) + 2Br^-(aq)$

C. $Br_2(l) + Cl_2(g) \rightarrow 2Br^-(aq) + 2Cl^-(aq)$

D. $2Br^-(aq) + 2Cl^-(aq) \rightarrow Br_2(l) + Cl_2(g)$

Question

The chemical reaction, $2AgCl(s) + H_2(g) \rightleftharpoons 2HCl(aq) + 2Ag(s)$ taking place in a galvanic cell is represented by the notation:

(A) Pt H_2(g), 1 bar | 1M KCl(aq) | AgCl(s) | Ag(s)

(B) Pt(s) | H_2(g), 1bar | 1M HCl(aq) || 1M Ag^+(aq) | Ag(s)

(C) Pt(s) | H_2(g), 1 bar | 1M HCl(aq) | AgCl(s) | Ag(s)

(D) Pt(s) | H_2(g), 1 bar | 1M HCl(aq) | Ag(s) | AgCl(s)

Question

How can an electrochemical cell be converted into an electrolytic cell?

A. Applying an external opposite potential greater than E^0_{cell}.

B. Reversing the flow of ions in salt bridge.

C. Applying an external opposite potential lower than E^0_{cell}.

D. Exchanging the electrodes at anode and cathode.

Question

The reaction;

$$\frac{1}{2}H_{2(g)} + AgCl_{(s)} \rightarrow H^{+}_{(aq)} + Cl^{-}_{(aq)} + Ag_{(s)}$$

occurs in which of the following galvanic cell:

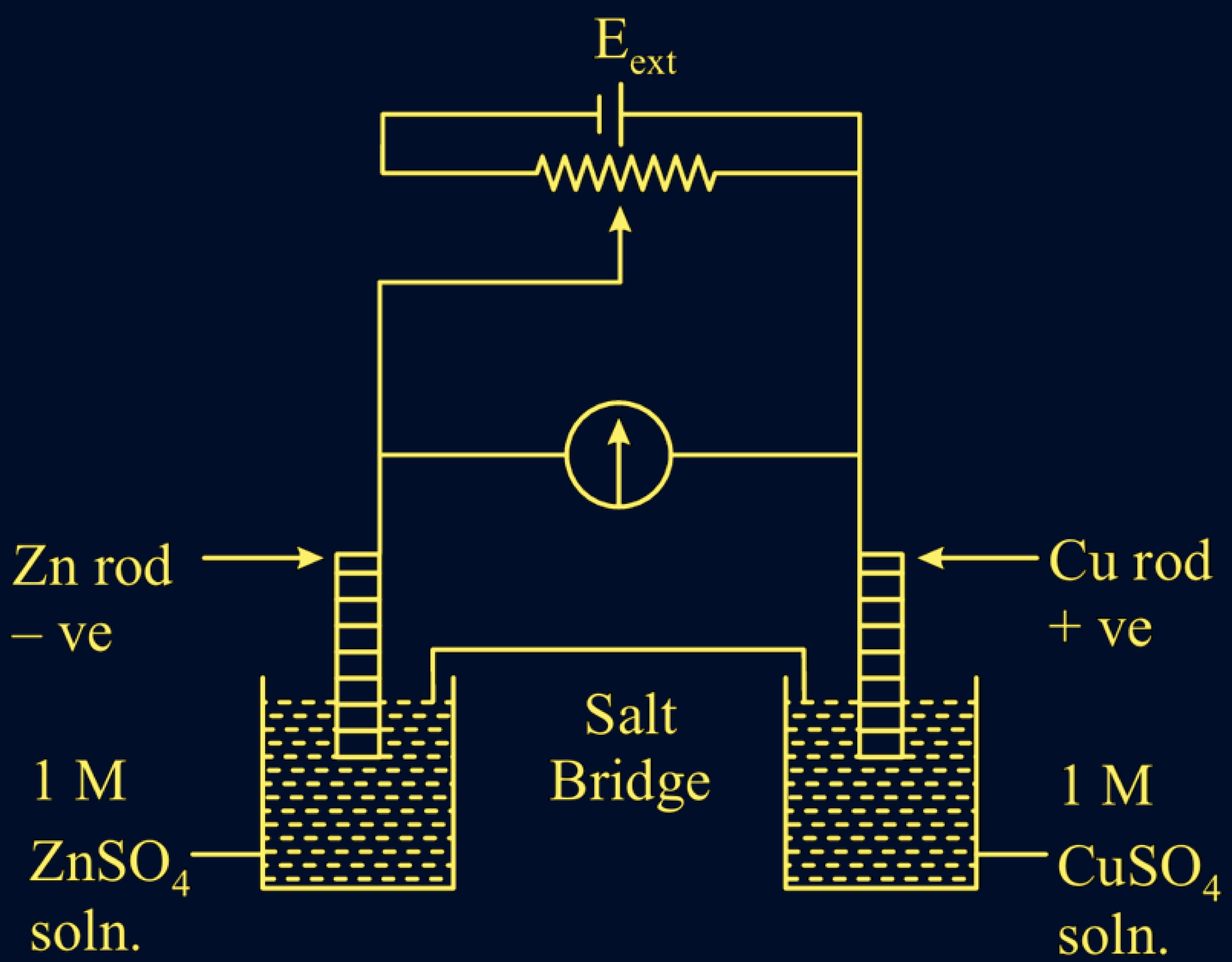
A. $Pt|H_{2(g)}|HCl_{(so\ ln.)}|AgCl_{(s)}|Ag$

B. $Pt|H_{2(g)}|HCl_{(so\ ln.)}|AgNO_{3(aq)}|Ag$

C. $Pt|H_{2(g)}|KCl_{(so\ ln.)}|AgCl_{(s)}|Ag$

D. $Ag|AgCl_{(s)}|KCl_{(so\ ln.)}|AgNO_{3(aq)}|Ag$

Question



$$E^0_{Cu^{2+}|Cu} = +0.34\ V$$
$$E^0_{Zn^{2+}|Zn} = -0.76\ V$$

Identify the incorrect statement from the option below for the above cell.

[4 Sept, 2020 (Shift-I)]

A. If $E_{ext} < 1.1$ V, Zn dissolves at anode and Cu deposits at cathode

B. If $E_{ext} = 1.1$ V, no flow of e^- or current occurs

C. If $E_{ext} > 1.1$ V, e^- flow from Cu to Zn

D. If $E_{ext} > 1.1$ V, Zn dissolve at Zn electrode and Cu deposits at Cu electrode

Question

The standard reduction potential at 298 K for the following half cells are given below:

$NO_3^- + 4H^+ + 3e^- \rightarrow NO_{(g)} + 2H_2O$ $\quad E^0 = 0.97V$

$V^{2+}(aq) + 2e^- \rightarrow V$ $\quad E^0 = -1.19V$

$Fe^{3+}(aq) + 3e^- \rightarrow Fe$ $\quad E^0 = -0.04V$

$Ag^+(aq) + e^- \rightarrow Ag(s)$ $\quad E^0 = 0.80V$

$Au^{3+}(aq) + 3e^- \rightarrow Au(s)$ $\quad E^0 = 1.40V$

The number of metal(s) which will be oxidized by NO_3^- in aqueous solution is

Question

The standard reduction potentials at 298 K for the following half cells are given below:

$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$, $E° = 1.33V$

$Fe^{3+}(aq) + 3e^- \rightarrow Fe$ $\quad E° = -0.04V$

$Ni^{2+}(aq) + 2e^- \rightarrow Ni$ $\quad E° = -0.25V$

$Ag^+(aq) + e^- \rightarrow Ag$ $\quad E° = 0.80V$

$Au^{3+}(aq) + 3e^- \rightarrow Au$ $\quad E° = 1.40V$

Consider the given electrochemical reactions, The number of metal(s) which will be oxidized be

$Cr_2O_7^{2-}$, in aqueous solution is ________. [09 April, 2024 (Shift-I)]

Question

The electrode potential measures the-

A. Tendency of the electrode to gain or lose electrons

B. Tendency of a cell reaction to occur

C. Difference in the ionisation potential of electrode and metal ion

D. Current carried by an electrode

Question

Standard reduction electrode potential of three metals X, Y and Z are –1.2V, +0.5V and –3.0V respectively. The reducing power of these metals will be

(A) $X > Y > Z$

(B) $Y > Z > X$

(C) $Y > X > Z$

(D) $Z > X > Y$

Question

Based on the data given below:

$E^{\circ}_{Cr_2O_7^{2-}/Cr^{3+}} = 1.33V$ $\quad E^{\circ}_{Cl_2/Cl^{(-)}} = 1.36V$

$E^{0}_{MnO_4^{-}/Mn^{2+}} = 1.51V$ $\quad E^{\circ}_{Cr^{3+}/Cr} = -0.74V$

the strongest reducing agent is:

(A) Mn^{2+}

(B) Cl^{-}

(C) MnO_4^{-}

(D) Cr

Question

The standard reduction potential values of some of the p-block ions are given below. Predict the one with the strongest oxidising capacity.

A. $E^{\Theta}_{Sn^{4+}/Sn^{2+}} = +1.15V$

B. $E^{\Theta}_{Pn^{4+}/Pb^{2+}} = +1.67V$

C. $E^{\Theta}_{Tl^{3+}/Tl} = +1.26V$

D. $E^{\Theta}_{Al^{3+}/Al} = -1.66V$

Question

Given that $E^{o}_{O_2/H_2O} = +1.23\ V$ [8 April, 2019 (Shift-I)]

$E^{e}_{S_2O_8^{2-}/SO_4^{2-}} = 2.05;$

$E^{o}_{Br_2/Br^-} = +1.09\ V;$

$E^{o}_{Au^{3+}/Au} = +1.4\ V$

The strongest oxidizing agent is:

(A) Br_2

(B) Au^{3+}

(C) $S_2O_8^{2-}$

(D) O_2

Question

Consider the following

$\mathbf{Zn^{2+} + 2e^- \longrightarrow Zn(s); E^\circ = -0.76\ V}$

$\mathbf{Ca^{2+} + 2e^- \longrightarrow Ca(s); E^\circ = -2.87\ V}$

$\mathbf{Mg^{2+} + 2e^- \longrightarrow Mg(s); E^\circ = -2.36\ V}$

$\mathbf{Ni^{2+} + 2e^- \longrightarrow Ni(s); E^\circ - 0.25\ V}$

The reducing power of the metals increases in the order

[10 Jan, 2019 (Shift-I)]

A. $Ca < Zn < Mg < Ni$

B. $Ni < Zn < Mg < Ca$

C. $Zn < Mg < Ni < Ca$

D. $Ca < Mg < Zn < Ni$

Question

The correct order of reduction potentials of the following pairs is

(A) Cl_2/Cl^- (B) I_2/I^-

(C) Ag^+/Ag (D) Na^+/Na (E) Li^+/Li [25 June, 2022 (Shift-II)]

A (A) > (C) > (B) > (D) > (E)

B (A) > (B) > (C) > (D) > (E)

C (A) > (C) > (B) > (E) > (D)

D (A) > (B) > (C) > (E) > (D)

Question

Standard electrode potentials of redox couples A^{2+}/A, B^{2+}/B, C/C^{2+} and D^{2+}/D are 0.3V, –0.5V, –0.75V and 0.9V respectively. Which of these is best oxidising agent and reducing agent respectively?

(A) D^{2+}/D and B^{2+}/B

(B) B^{2+}/B and D^{2+}/D

(C) D^{2+}/D and C^{2+}/C

(D) C^{2+}/C and D^{2+}/D

Question

In the cell $Cu \mid Cu^{2+} \parallel Ag^{+} \mid Ag$ which is not applicable?

(A) Ag electrode is negative electrode.

(B) Cu electrode is negative electrode.

(C) Ag^{+} is reduced to Ag.

(D) Cu is oxidised to Cu^{2+}.

Question

Which of the following statements is correct with respect to both electrolytic cell and Galvanic cell?

A. In both cells, anode is shown by +ve sign.

B. In both cells, cathode is shown by –ve sign.

C. In both cells, reduction reaction takes place at the cathode.

D. In both cells, oxidation reaction takes place at the cathode.

Question

Assertion (A): Anode is the electrode at which oxidation occurs and cathode is the electrode at which reduction occurs.

Reason (R): Anode and cathode in electrochemical cells and electrolytic cells have opposite polarity.

A. Both Assertion (A) and Reason (R) are true and the Reason (R) is the correct explanation of the Assertion (A).

B. Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of the Assertion (A).

C. Assertion (A) is true and Reason (R) is false.

D. Assertion (A) is false and Reason (R) is true.

Question

Statement-I: Galvanic cell converts the chemical energy liberated during the reaction to electrical energy. The chemical reaction will be a spontaneous reaction.

Statement-II: In electrolytic cells, electrical energy is being used to carry out chemical reactions (chemical energy). Here, the electric current drives a non-spontaneous reaction.

(A) Statement-I and Statement-II both are correct.

(B) Statement-I is correct but Statement-II is incorrect.

(C) Statement-I is incorrect but Statement-II is correct.

(D) Statement-I and Statement-II both are incorrect.

Question

In a galvanic cell, if external potential is given and increases slowly then:

A. reaction stops.

B. instantly reverse reaction starts.

C. when external potential does not become 1.1V reaction will continue.

D. no reaction changes occurred in cell.

Question

In an electrochemical cell, the electrode having a higher reduction potential will act as

A. Salt

B. Electrolyte

C. Anode

D. Cathode

Question

Standard reduction potentials of the half reactions are given below:

$F_2(g) + 2e^- \rightarrow 2F^-(aq)$; $E° = +2.85$ V

$Cl_2(g) + 2e^- \rightarrow 2Cl^-(aq)$; $E° = +1.36$ V

$Br_2(l) + 2e^- \rightarrow 2Br^-(aq)$; $E° = +1.06$ V

$I_2(s) + 2e^- \rightarrow 2I^-(aq)$; $E° = +0.53$ V

The strongest oxidizing and reducing agents respectively are

A. F_2 and I^-

B. Br_2 and Cl^-

C. Cl_2 and Br^-

D. Cl_2 and I_2

Question

The reduction potential values are given below:
$Al^{3+}/Al = -1.67$ volt, $Mg^{2+}/Mg = -2.34$ volt
$Cu^{2+}/Cu = +0.34$ volt, $I_2/2I^- = +0.53$ volt
Which one is best reducing agent:

A. Al

B. Mg

C. Cu

D. I_2

Question

The standard potential, E°, for the half reaction areas

Zn $\quad$ $Zn^{2+} + 2e^-$, E° = +0.76 V

Fe $\quad$ $Fe^{2+} + 2e^-$, E° = +0.41 V

The EMF for the cell reaction is.

$Fe^{+2} + Zn \rightarrow Zn^{+2} + Fe$

(A) –0.35 V

(B) +0.35 V

(C) +1.17 V

(D) –1.17 V

Question

The standard electrodes potentials for the reactions

$Ag^+ (aq.) + e^- \rightarrow Ag(s)$,

$Sn^{2+} (aq.) + 2e^- \rightarrow Sn(s)$,

at 25°C are 0.80 volts and –0.14 volt, respectively. The emf of the cell.

$Sn \mid Sn^{2+} (1\ M) \parallel Ag^+ (1\ M) \mid Ag$ is:

(A) 0.66 volt

(B) 0.80 volt

(C) 1.08 volt

(D) 0.94 volt

Question

For the electro chemical cell

$M|M^{2+}||X|X^{2-}$

If $E^0_{(M^{2+}/M)} = 0.46$ V and $E^0_{(X/X^{2-})} = 0.34$ V

Which of the following is correct?

(A) $E_{cell} = -0.80$ V

(B) $M + X \rightarrow M^2 + X^{2-}$ is a spontaneous reaction

(C) $M^{2+} + X^{2-} \rightarrow M + X$ is a spontaneous reaction

(D) $E_{cell} = 0.80$ V

Question

E^o for the half cell $Zn^{2+} || Zn$ is –0.76 emf of the cell $Zn | Zn^{2+} (1\ M) || 2H^+ (1\ M) | H_2 (1\ atm)$ is.

(A) –0.76 V

(B) +0.76 V

(C) –0.38 V

(D) +0.38 V

Question

From the following E^o values of half cells,

(i) $A^{3-} \rightarrow A^{-2} + e$; $E^o = 1.5$ V

(ii) $B^+ + e \rightarrow B$; $E^o = 0.5$ V

(iii) $C^{2-} + e \rightarrow C^+ + e$; $E^o = 0.5$ V

(iv) $D \rightarrow D^{2+} + 2e$; $E^o = -1.15$ V

What combination of two half cells would result in a cell with the largest potential?

A (i) and (iii)

B (i) and (iv)

C (ii) and (iv)

D (iii) and (iv)

Question

EMF of cell Ni | Ni^{+2}(1.0M) || Au^{+3} (1.0M) | Au (Where E^o for Ni^{+2}/Ni is-0.25V; E^o for Au^{+3}/Au is 1.50 V) is.

A. +1.25 V

B. –1.75 V

C. +1.75 V

D. +4.0 V

Question

For a $Mg|Mg^{2+}(aq)||Ag^{+}(aq)|Ag$ the correct Nernst Equation is:

[29 Jan, 2025 (Shift-I)]

(A) $E_{cell} = E^{\circ}_{cell} + \frac{RT}{2F} \ln \frac{[Ag^{+}]^2}{[Mg^{2+}]}$

(B) $E_{cell} = E^{\circ}_{cell} - \frac{RT}{2F} \ln \frac{[Ag^{+}]}{[Mg^{2+}]}$

(C) $E_{cell} = E^{\circ}_{cell} - \frac{RT}{2F} \ln \frac{[Mg^{2+}]}{[Ag^{+}]}$

(D) $E_{cell} = E^{\circ}_{cell} - \frac{RT}{2F} \ln \frac{[Ag^{+}]^2}{[Mg^{2+}]}$

Question

Consider the cell [30 Jan, 2023 (Shift-I)]

$$Pt_{(s)}|H_2(g, 1\ atm)|H^+(aq, 1M)||Fe^{3+}(aq), Fe^{2+}(aq) \mid Pt(s)$$

When the potential of the cell is 0.712 V at 298 K, the ratio $[Fe^{2+}]/[Fe^{3+}]$ is . (Nearest integer)

Given: $Fe^{3+} + e^- = Fe^{2+}, E^0_{Fe^{3+},Fe^{2+}|Pt} = 0.771$

$$\frac{2.303RT}{F} = 0.06\ V$$

Question

Consider the following equations for a cell reaction

$A + B \rightleftharpoons C + D$;

$E^o = x$ volt, $K_{eq} = E_1$

$2A + 2B \rightleftharpoons 2C + 2D$;

$E^o = y$ volt, $K_{eq} = K_2$

then:

A. $x = y, K_1 = K_2$

B. $x = 2y, K_1 = 2K_2$

C. $x = y, K_1{}^2 = K_2$

D. $x^2 = y, K_1{}^2 = K_2$

Question

The standard electrode potential of zinc ions is 0.76 V. What will be the potential of a 2M solution at 300K?

A. 0.83 V

B. 0.76 V

C. 0.23 V

D. 0.98 V

Question

The standard oxidation potential E^o for the half reaction are.

$Zn \rightarrow Zn^{+2} + 2e^-$ $\qquad E^o = 0.76\ V$

$Ag \rightarrow Ag^+ + e^-$ $\qquad E^o = -0.77\ V$

The standard EMF of the cell

$Ag^+ + Zn \rightarrow Zn^{+2} + Ag$ is

A) 1.53 V

B) –1.53 V

C) –0.01 V

D) 0.01 V

Question

Given electrode potentials:

$Fe^{3+} + e \rightarrow Fe^{2+}$; $E^o = 0.771$ volts

$I_2 + 2e \rightarrow 2I^-$; $E^o = 0.536$ volts

E^o cell for the cell reaction

$2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$ is

A. $(2 \times 0.771 - 0.536) = 1.006$ volts

B. $(0.771 - 0.5 \times 0.536) = 0.503$ volts

C. $0.771 - 0.536 = 0.235$ volts

D. $0.536 - 0.771 = -0.235$ volts

Question

For the reaction $Zn(s) + Cu^{2+} \rightarrow Zn^{2+}(aq) + Cu(s)$, Nernst equation at 25°C is:

(A) $E = E^{o} + \dfrac{0.059}{2} \log \dfrac{\left[Zn^{2+}_{(aq)}\right]}{\left[Cu^{2+}_{(aq)}\right]}$

(B) $E = E^{o} + \dfrac{0.059}{2} \log \dfrac{\left[Zn^{2+}_{(aq)}\right]}{\left[Zn^{2+}_{(aq)}\right]}$

(C) $E = E^{o} + \dfrac{0.059}{2} \log \dfrac{\left[Cu^{2+}_{(aq)}\right]}{\left[Zn^{2+}_{(aq)}\right]}$

(D) $E = E^{o} + \dfrac{0.059}{2} \log \dfrac{\left[Cu^{2+}_{(aq)}\right]}{\left[Cu^{2+}_{(aq)}\right]}$

Question

The Nernst equation given dependence of electrode oxidation potential on concentration is:

A) $E = E^{o} + \frac{2.303RT}{nF}\log[M^{+n}]$

B) $E = E^{o} - \frac{2.303RT}{nF}\log\frac{[M^{n+}]}{[M]}$

C) $E = E^{o} - \frac{2.303RT}{nF}\log[M^{n+}]$

D) $E = E^{o} + \frac{2.303RT}{nF}\log\frac{[M]}{[M^{n+}]}$

Question

Calculate the electrode potential (in volt) of the following electrode at 25°C.
Ni^{+2} (0.1M)|Ni(s)

$$\left[E^{o}_{(Ni^{2+}|Ni)} = -0.25V, \frac{2.303RT}{F} = 0.06\right]$$

(A) –0.28 V

(B) –0.34 V

(C) –0.82 V

(D) –0.22 V

Question

The standard reduction potentials of $Zn^{2+} | Zn$ and $Cu^{2+} | Cu$ are –0.76V and +0.34 V respectively. What is the cell e.m.f (inV) of the following cell?

$Zn | Zn^{2+}$ (0.05M) || Cu2+ (0.005M) | Cu

$$\left(\frac{RT}{F} = 0.059\right)$$

(A) 1.1295

(B) 1.0705

(C) 1.1

(D) 1.041

Question

Find the emf of the cell in which the following reaction takes place at 298 K

$Ni(s) + 2Ag^{+}$ (0.001 M) $\rightarrow Ni^{2+}$ (0.001 M) $+ 2Ag(s)$

(Given that $E^{\circ}_{cell} = 1.05\ V, \frac{2.303RT}{F} = 0.059$ at 298 K)

(A) 1.05 V

(B) 1.0385 V

(C) 1.385 V

(D) 0.9615 V

Question

The electrode potential of the following half cell at 298 K.

$X|X^{2+}(0.001M)||Y^{2+}(0.01M)|Y$ is $\times 10^{-2}$ V (Nearest integer).

Given: $E^0_{X^{2+}|X} = -2.36$ V $\quad\quad E^0_{Y^{2+}|Y} = +0.36$ V

$$\frac{2.303RT}{F} = 0.06 \text{ V}$$

Question

For the cell $Zn | Zn^{2+} || Cu^{2+} | Cu$, if the concentration of both, Zn^{2+} and Cu^{2+} ions are doubled, the EMF of the cell.

A. Doubles

B. Reduces to half

C. Remains same

D. Becomes zero

Question

The emf of the cell $Tl(s) \mid Tl^{+}$ (0.0001 M) $\|$ Cu^{2+} (0.01 M) $\mid Cu(s)$ is 0.83 V
The emf of this cell will be increased by:

A. Increasing the concentration of Cu^{+2} ions

B. Decreasing the concentration of Tl^{+}

C. Increasing the concentration of both

D. (A) and (B) both

Question

Consider the following four electrodes;

A = Cu^{2+} (0.0001M)|Cu(s)

B = Cu^{2+} (0.1M) Cu(s)

C = Cu^{2+} (0.01M)|Cu(s)

D = Cu^{2+} (0.001M)|Cu(s)

If the standard reduction potential of Cu^{+2}/Cu is +0.34V, then the reduction potentials (in volts) of the above electrodes follow the order:

(A) A > D > C > B

(B) B > C > D > A

(C) C > D > B > A

(D) A > B > C > D

Question

An oxidation-reduction reaction in which 3 electrons are transferred has ΔG° of 17.37 kJ mol^{-1} at $25^\circ C$. The value of E^0_{cell} (in V) is ________ $\times 10^{-2}$.
$(1\ F = 96,500\ C\ mol^{-1})$ **[8 Sept, 2020 (Shift-I)]**

Question

The logarithm of equilibrium constant for the reaction $Pd^{2+} + 4Cl^- \rightleftharpoons PdCl_4^{2-}$

Given: $\frac{2.303RT}{F} = 0.06\ V$

$$Pd^{2+}_{(aq)} + 2e^- \rightleftharpoons Pd(s) \quad E^\circ = 0.83\ V$$

$$PdCl_4^{2-}(aq) + 2e^- \rightleftharpoons Pd(s) + 4Cl^-(aq)\ E^\circ = 0.65\ V$$

Question

The equilibrium constant for the reaction $Zn(s) + Sn^{2+}(aq) \rightleftharpoons Zn^{2+}(aq) + Sn(s)$ is 1×10^{20} at $298\ K$. The magnitude of standard electrode potential of Sn/Sn^{2+} is $\times 10^{-2}$ V. if $E^{\circ}_{Zn^{2+}/Zn} = -0.76\ V$

Given: $\frac{2.303RT}{F} = 0.059\ V$

Question

Standard electrode potentials for a few half cells are mentioned below:

$$E^\circ_{Cu^{2+}/Cu} = 0.34\ V,\ E^\circ_{Zn^{2+}/Zn} = -0.76\ V$$

$$E^\circ_{Ag^{+}/Ag} = 0.80\ V,\ E^\circ_{Mg^{2+}/Mg} = -2.37\ V$$

Which one of the following cells gives the most negative value of ΔG°?

A. $Zn|Zn^{2+}(1M)||Ag^{+}(1M)|Ag$

B. $Cu|Cu^{2+}(1M)||Ag^{+}(1M)|Ag$

C. $Zn|Zn^{2+}(1M)||Mg^{2+}(1M)|Mg$

D. $Ag|Ag^{+}(1M)||Mg^{2+}(1M)|Mg$

Question

The EMF of the cell: Zn | Zn^{2+} (0.01 M) || Fe^{2+} (0.001 M) | FE at 298 K is 0.2905 V, then the value of equilibrium constant for the cell reaction is.

A. $e^{\frac{0.32}{0.0295}}$

B. $10^{\frac{0.32}{0.0295}}$

C. $10^{\frac{0.26}{0.0295}}$

D. $10^{\frac{0.32}{0.0591}}$

Question

The value of equilibrium constant for feasible cell reaction is.

A. <1

B. zero

C. $=1$

D. >1

Question

The oxidation potential of hydrogen electrode at pH = 10 and P_{H_2} = 1 atm is

A. 0.51 V

B. 0.00 V

C. +0.59 V

D. 0.059

Question

Given, $E^{o}_{I_2/I^-}(1M) = +0.54V$,
and $E^{o}_{Br_2/Br^-}(1M) = 1.09V$
On this basis the feasible reaction is

A. $2Br^- + I_2 \rightarrow Br_2 + 2I^-$

B. $I_2 + Br_2 \rightarrow 2I^- + 2Br^-$

C. $2I^- + Br_2 \rightarrow I_2 + 2Br^-$

D. $2I^- + 2Br^- \rightarrow I_2 + Br_2$

Question

For the cell,

$Cu \mid Cu^{2+} \parallel Ag^{+} \mid Ag$, $E^{o}_{cell} = +0.46$ V

If concentration of Cu^{2+} ions is doubled, then E^{o}_{cell} will be:

A. halved

B. doubled

C. four times

D. remains the same

Question

The emf of the cell

Ni/Ni^{+2} (1.0 M) || Au^{+3} (0.1 M)/Au

[E° for $Ni^{+2}/Ni = -0.25$, E° for $Au^{+3}/Au = 1.50$ V] is given as:

A. 1.25 V

B. –1.75 V

C. 1.75 V

D. 1.73 V

Question

The hydrogen electrode is dipped in a solution of pH = 3 at 25°C. The oxidation potential of the electrode would be:

A) 0.177 V

B) –0.177 V

C) 0.087 V

D) 1.08 V

Question

For a reaction $A(s) + 2B^{+} \rightarrow 2B + A^{2+}$, K_c has been found to be 10^{12} the std. emf of the cell is-

(A) 0.354 V

(B) 0.708 V

(C) 0.534 V

(D) 0.453 V

Question

The equilibrium constant for a cell reaction:
$Cu(g) + 2Ag^{+}(aq) \rightarrow Cu^{+2}(aq) + 2Ag(s)$ is
4×10^{16}. Find E_{cell}^{o} for the cell reaction.

A) 0.63 V

B) 0.49 V

C) 1.23 V

D) 3.24 V

Question

Given standard electrode potentials

$Fe^{++} + 2e^{-}$ Fe ; $E^{o} = -0.440$ V

$Fe^{+++} + 3e^{-}$ Fe ; $E^{o} = -0.0036$ V

The standard electrode potential (E^{o}) for

$Fe^{+++} + e^{-}$ Fe^{+}+ is

(A) –0.476 V

(B) –0.404 V

(C) +0.404 V

(D) +0.772 V

Question

When two half-cells of electrode potential of E_1 and E_2 are combined to form a cell of electrode potential E_3, then
(when n_1, n_2 and n_3 are no. of electrons exchanged in first, second and combined half-cells):

A. $E_3 = E_2 - E_1$

B. $E_3 = \frac{E_1 n_1 + E_2 n_2}{n_3}$

C. $E_3 = \frac{E_1 n_1 - E_2 n_2}{{n_3}^2}$

D. $E_3 = E_1 + E_2$

Question

Standard electrode potential for

$Cu^{+2} + e^- \rightarrow Cu^+$

$Cu^+ + e^- \rightarrow Cu$

are 0.15 V and 0.50 V respectively.

The value of $E^o_{Cu^{+2}/cu}$ will be.

A 0.150 V

B 0.50 V

C 0.325 V

D 0.650 V

Question

E^o for the half cell $Zn^{2+} || Zn$ is –0.76 emf of the cell $Zn | Zn^{2+}$ (1 M) || $2H^+$ (1 M) | H_2 (1 atm) is.

(A) –0.76 V

(B) +0.76 V

(C) –0.38 V

(D) +0.38 V

Question

Standard electrode potential (reduction potential) becomes equal to cell potential, when

A. either anode or cathode is the hydrogen electrode

B. anode is the hydrogen electrode

C. cathode is the hydrogen electrode

D. None of the above

Question

$Pt(s) \mid H_2(g, 1bar) \mid H^+ (aq, 1M) \parallel Cu^{2+} (aq, 1M) \mid Cu$

The measured emf of the cell is 0.34 V means

A. Cu^{2+} ions get reduced easily than H^+ ion

B. hydrogen ions cannot oxidise Cu

C. copper does not dissolve in HCl

D. All of the above

Question

A standard hydrogen electrode has a zero potential because:

A. hydrogen can be most easily oxidised.

B. the electrode potential is assumed to be zero.

C. hydrogen has only one electron.

D. hydrogen is the lightest element.

Question

Statement-I: Absolute value of E_{red}^{o} of an electrode cannot be determined.
Statement-II: Neither oxidation nor reduction can take place alone.

A. Statement-I and Statement-II both are correct.

B. Statement-I is correct but Statement-II is incorrect.

C. Statement-I is incorrect but Statement-II is correct.

D. Statement-I and Statement-II both are incorrect.

Question

The potential of the cell containing two hydrogen electrodes as represented below:

$Pt : \frac{1}{2} H_2(g) \mid H^+ (10^{-8}\ M) \parallel H^+ (0.001\ M) \mid \frac{1}{2} H_2(g) \mid Pt$, is

(A) 0.296 V

(B) –0.295 V

(C) 0.13 V

(D) –0.13 V

Question

Assuming that hydrogen behaves as an ideal gas, what is the EMF of the cell at 25°C if P_1 = 600 mm and P_2 = 420 mm:
Pt | $H_2(P_1)$ | HCl || $H_2(P_2)$ | Pt?
[Given: 2.303 RT/F = 0.06, log 7 = 0.85]

(A) –0.0045 V

(B) –0.0 V

(C) +0.0045 V

(D) +0.0015 V

Question

The reduction potential of hydrogen half-cell will be negative if

(A) $p(H_2) = 1$ atm and $[H^+] = 2.0$ M

(B) $p(H_2) = 1$ atm and $[H^+] = 1.0$ M

(C) $p(H_2) = 2$ atm and $[H^+] = 1.0$ M

(D) $p(H_2) = 2$ atm and $[H^+] = 2.0$ M

Question

$Pt(s)H_2(g)(1bar)|H^+(aq)(1M)||M^{3+}(aq), M^+(aq)|Pt(s)$

The E_{cell} for the given cell is 0.1115 V at 298 K

When $\frac{[M^+(aq)]}{[M^{3+}(aq)]} = 10^a$

The value of a is

Given: $E^\theta_{M^{3+}/M^+} = 0.2\ V$

$$E^\theta_{M^{3+}/M^-} = \frac{2.303RT}{F} = 0.059\ V$$

Question

Electrolytes conducts electricity due to

A. Flow of ions

B. Flow of electrons

C. Both

D. None

Question

Cell constant has the unit

A. cm

B. cm^{-1}

C. cm^2

D. $cm\ sec^{-1}$

Question

The resistance of a conductivity cell containing 0.001M KCl solution at 298 K is 1500 Ω. What is the cell constant if conductivity of 0.001M KCl solution at 298 K is 0.146×10^{-3} S cm^{-1}.

Question

Cell constant of an electrolytic solution is 0.5 cm^{-1} and resistivity 54 ohm-cm. Find conductance of the electrolytic solution.

A. $\frac{1}{36}$

B. $\frac{1}{56}$

C. $\frac{1}{97}$

D. $\frac{1}{27}$

Question

The resistance of 0.05 N solution of an electrolyte was found to be 420 ohm at 298 K. Its conductance is:

A. 2.4×10^{-3} ohm^{-1}

B. 8.4×10^{-3} ohm^{-1}

C. 5.6×10^{-4} ohm^{-1}

D. 7.2×10^{-3} ohm^{-1}

Question

Correct expression for conductance of an electrolyte whose cell constant is 'a' resistivity 'X' is:

(A) $\frac{2}{Xa}$

(B) $\frac{1}{X}$

(C) $\frac{a}{X}$

(D) $\frac{1}{X.a}$

Question

The electrical resistance of a column of 0.05 mol L^{-1} NaOH solution of diameter 1 cm and length 50 cm is 5.55×10^3 ohm. Calculate its resistivity, conductivity and molar conductivity.

Question

The conductivity of 0.20 M solution of KCl at 298 K is 0.0248 S cm^{-1}. Calculate its molar conductivity.

Question

Resistance of a conductivity cell filled with 0.1 mol L^{-1} KCl solution is 100 Ω. If the resistance of the same cell when filled with 0.02 mol L^{-1} KCl solution is 520 Ω, calculate the conductivity and molar conductivity of 0.02 mol L^{-1} KCl solution. The conductivity of 0.1 mol L^{-1} KCl solution is 1.29 S/m.

Question

The resistance of 0.5 M solution of an electrolyte in a cell was found to be 50 Ω. If the electrodes in the cell are 2.2 cm apart and have an area of 4.4 cm^2 then the molar conductivity (in $S\ m^2\ mol^{-1}$) of the solution is:

(A) 0.2

(B) 0.02

(C) 0.002

(D) None of these

Question

The specific conductance of a solution is 0.3568 ohm^{-1} cm^{-1}. When placed in a cell the conductance is 0.0268 ohm^{-1}. The cell constant is

A. 1.331 cm^{-1}

B. 13.31 cm^{-1}

C. 0.665 cm^{-1}

D. 6.65 cm^{-1}

Question

Match the List-I with List-II and List-III.

A. (a – q, u), (b – s, v), (c – d, w), (d – r, x)

B. (a – q, u), (b – s, v), (c – p, w), (d – r, x)

C. (a – b, u), (b – s, v), (c – p, w), (d – r, x)

D. (a – p, u), (b – s, v), (c – p, w), (d – r, x)

List-I (Quantity)	List-II (Symbol)	List-III (Unit)
(1) Conductivity	(p) Λ_m	(u) mho cm^{-1}
(2) Cell constant	(q) K	(v) cm^{-1}
(3) Molar conductance	(r) Λ_{eq}	(w) Ohm^{-1} cr mol^{-1}
(4) Equivalent conductance	(s) l/A	(x) ohm^{-1} cm^2 ec 1

Question

The specific conductance's in $ohm^{-1}\ cm^{-1}$ of four electrolytes P, Q, R and S are given in brackets:

$P(5.0 \times 10^{-5})$ $Q(7.0 \times 10^{-8})$

$R(1.0 \times 10^{-10})$ $S(9.2 \times 10^{-3})$

The one that offers highest resistance to the passage of electric current is

(A) P

(B) S

(C) R

(D) Q

Question

When equilibrium is reached, the potential difference between the two electrodes is:

A. <1

B. >1

C. 0

D. None

Question

If E_{cell}^{o} for a given reaction has a positive value, then which of the following is correct?

A. $\Delta G^{o} > 0, K_C < 1$

B. $\Delta G^{o} > 0, K_C > 1$

C. $\Delta G^{o} < 0, K_C > 1$

D. $\Delta G^{o} < 0, K_C < 1$

Question

For a cell involving one electron, E_{cell}^{o} = 0.59 V at 298 K, the equilibrium constant for the reaction is.

$$\frac{2.303RT}{F} = 0.059V$$

(A) 10^{30}

(B) 10^{2}

(C) 10^{5}

(D) 10^{10}

Question

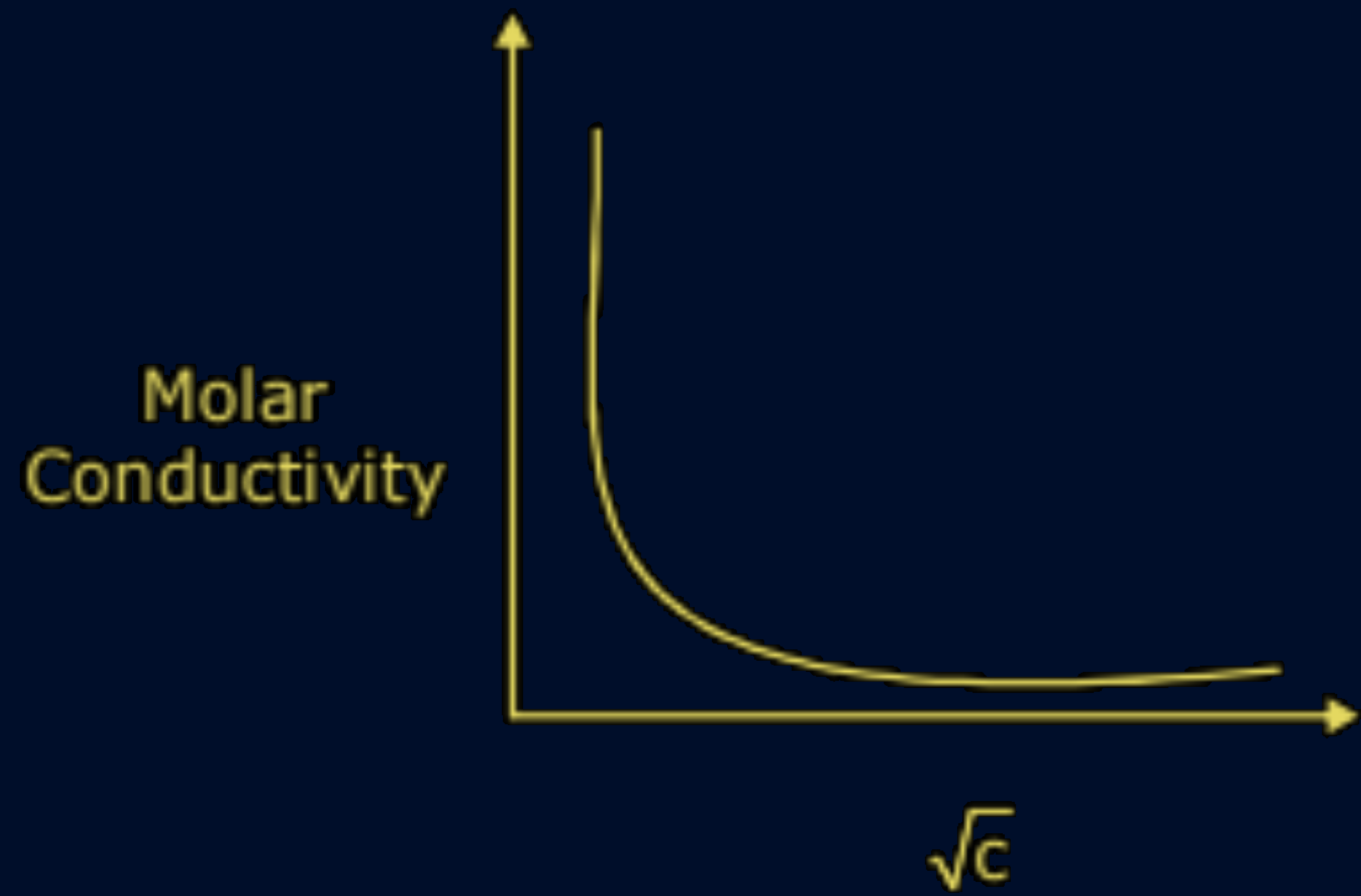
The variation of molar conductivity with concentration of an electrolyte (X) in aqueous solution is shown in the given figure.
The electrolyte X is

A) HCl

B) NaCl

C) KNO_3

D) CH_3COOH



Question

Ionic conductivity of Al^{+3} and SO_4^{-2} ions at infinite dilution are 189 $S\ cm^2\ mol^{-1}$ and 160 $S\ cm^2\ mol^{-1}$ respectively. Find the molar conductivity at infinite dilution $Al_2(SO_4)_3$

A. 192 $S\ cm^2\ mol^{-1}$

B. 858 $S\ cm^2\ mol^{-1}$

C. 857 $S\ cm^2\ mol^{-1}$

D. 143 $S\ cm^2\ mol^{-1}$

Question

Λ^0m for NaCl, HCl and NaAc are 126.4, 425.9 and 91.0 S cm^2 mol^{-1} respectively. Calculate Λ^0 for HAc.

Question

Molar conductance's of $BaCl_2$, H_2SO_4 and HCl at infinite dilutions are x_1, x_2 and x_3, respectively. Equivalent conductance of $BaSO_4$ at infinite dilution will be

A. $\frac{[x_1 + x_2 - x_3]}{2}$

B. $\frac{[x_1 - x_2 - x_3]}{2}$

C. $2(x_1 + x_2 - 2x_3)$

D. $\frac{[x_1 + x_2 - 2x_3]}{2}$

Question

The molar conductivity of 0.025 mol L^{-1} methanoic acid is 46.1 S cm^2 mol^{-1}. Calculate its degree of dissociation.

Given λ_0 (H^+) = 349.6 S cm^2 mol^{-1} and λ_0 ($HCOO^-$) = 54.6 S cm^2 mol^{-1}.

Question

The conductivity of 0.001028 mol L^{-1} acetic acid is 4.95×10^{-5} S cm^{-1}. Calculate its dissociation constant if $\Lambda^0 m$ for acetic acid is 390.5 S cm^2 mol^{-1}

Question

The conductivity of a saturated solution of $BaSO_4$ is 3.06×10^{-6} ohm^{-1} cm^{-1} and its molar conductance is 1.53 ohm^{-1} cm^2 mol^{-1}. The K_{sp} of $BaSO_4$ will be.

A) 4×10^{-12}

B) 2.5×10^{-9}

C) 2.5×10^{-13}

D) 4×10^{-6}

Question

The conductivity of a saturated solution of $BaSO_4$ is 3.06×10^{-6} ohm^{-1} cm^{-1} and its molar conductance is 1.53 ohm^{-1} cm^{2} mol^{-1}. The K_{sp} of $BaSO_4$ will be.

A) 4×10^{-12}

B) 2.5×10^{-9}

C) 2.5×10^{-13}

D) 4×10^{-6}

Question

Conductivity of a saturated solution of a sparingly soluble salt AB at 298K is $1.85 \times 10^{-5} Sm^{-1}$. Solubility product of the salt AB at 298K is:
[Given; $\Lambda_m^0(AB) = 140 \times 10^{-4} Sm^{-2}\ mol^{-1}$]

A 5.7×10^{-12}

B 1.32×10^{-12}

C 7.5×10^{-12}

D 1.74×10^{-12}

Question

Kohlrausch's law states that at

A. infinite dilution, each ion makes definite contribution to conductance of ar
electrolyte whatever be the nature of the other ion of the electrolyte

B. infinite dilution, each ion makes definite contribution to equivalent conductanc
of an electrolyte, whatever be the nature of the other ion of the electrolyte

C. finite dilution, each ion makes definite contribution to equivalent conductance o
an electrolyte, whatever be the nature of the other ion of the electrolyte

D. infinite dilution each ion makes definite contribution to equivalent conductanc
of an electrolyte depending on the nature of the other ion of the electrolyte.

Question

Kohlrausch law of independent migration of ion is applicable to

A. Only to weak electrolytes at a certain concentration.

B. Only to strong electrolytes at all concentrations

C. To both - strong and weak electrolytes

D. To non-electrolytes

Question

Limiting molar ionic conductivities of a univalent electrolyte are 57 and 73. The limiting molar conductivity of the solution will be:

A) $130 \text{ s cm}^2 \text{ mol}^{-1}$

B) $65 \text{ s cm}^2 \text{ mol}^{-1}$

C) $260 \text{ s cm}^2 \text{ mol}^{-1}$

D) $187 \text{ s cm}^2 \text{ mol}^{-1}$

Question

The correct order of equivalent conductance at infinite dilution in water at room temperature for H^+, K^+, CH_3COO^- and HO^- ions is:

(A) $HO^- > H^+ > K^+ > CH_3COO^-$

(B) $H^+ > HO^- > K^+ > CH_3COO^-$

(C) $H^+ > K^+ > HO^- > CH_3COO^-$

(D) $H^+ > K^+ > CH_3COO^- > HO^-$

Question

The sequence of ionic mobility in aqueous solutions is:

A) $K^+ > Na^+ > Rb^+ > Cs^+$

B) $Cs^+ > Rb^+ > K^+ > Na^+$

C) $Rb^+ > K^+ > Cs^+ > Na^+$

D) $Na^+ > K^+ > Rb^+ > Cs^+$

Question

The equivalent conductance of Ba^{2+} and Cl^{-} are respectively 127 and 76 $ohm^{-1}cm^{2}eq^{-1}$ at infinite dilution. What will be the equivalent conductance of $BaCl_2$ at infinite dilution?

A. 139.5

B. 101.5

C. 203

D. 279

Question

Equivalent conductance of 1 M of CH_3COOH is 10 $ohm^{-1}\ cm^2 eq^{-1}$ and that at infinite dilution is 200 $ohm^{-1}\ cm^2 eq^{-1}$. Hence the % ionization of CH_3COOH is:

A. 5%

B. 2%

C. 4%

D. 1%

Question

At 18°C, the conductance of H^+ and CH_3COO^- at infinite dilution are 315 and 35 mho $cm^2 eq^{-1}$ respectively. The equivalent conductivity of CH_3COOH at infinite dilution is ______(mho cm^2 eq^{-1})

A. 350

B. 280

C. 30

D. 315

Question

The current in a given wire is 1.8 A. The number of coulombs that flow in 1.36 minutes will be.

A 100 C

B 147 C

C 247 C

D 347 C

Question

What current strength in ampere will be required to liberate 10 g of chlorine from sodium chloride solution in one hour?

A. 7.55 ampere

B. 7.15 ampere

C. 7.50 ampere

D. 7.45 ampere

Question

An electric current of 100 amperes is passed through a molten liquid of sodium chloride for 5 hours. Calculate the volume of chlorine gas liberated at the electrode at NTP.

A) 210.5 L

B) 200 L

C) 211 L

D) 208.91 L

Question

Number of faraday required to liberate 8g of H_2 is-

A. 8

B. 16

C. 4

D. 2

Question

Number of electrons involved in the electrodeposition of 63.5 gm. of Cu from a solution of $CuSO_4$ is:

A. 6.022×10^{23}

B. 3.011×10^{23}

C. 12.044×10^{23}

D. 6.022×10^{22}

Question

The amount of electricity that can deposit 108 g. of silver from silver nitrate solution is:

A 1 ampere

B 1 coulomb

C 1 Faraday

D 2 ampere

Question

If 0.5 amp current is passed through acidified silver nitrate solution for 10 minutes. The mass of silver deposited on cathode is:
(Molar mass of silver $= 108$)

A. 0.235 g

B. 0.336 g

C. 0.536 g

D. 0.636 g

Question

A solution of $Ni(NO_3)_2$ is electrolysed between platinum electrodes using 0.1 Faraday electricity. How many mole of Ni will be deposited at the cathode?

(A) 0.05

(B) 0.20

(C) 0.15

(D) 0.10

Question

The same amount of electricity was passed through two cells containing molten Al_2O_3 and molten NaCl. If 1.8 g of Al were liberated in one cell, the amount of Na liberated in the other cell is:

(A) 4.6 g

(B) 2.3 g

(C) 6.4 g

(D) 3.2 g

Question

Three Faradays of electricity are passed through molten Al_2O_3, aqueous solution of $CuSO_4$ and molten NaCl taken in three different electrolytic cells. The amount of Al, Cu and Na deposited at the cathodes will be in the ratio of-

(A) 1 mole : 2 mole : 3 mole

(B) 1 mole : 1.5 mole : 3 mole

(C) 3 mole : 2 mole : 1 mole

(D) 1 mole : 1.5 mole : 2 mole

Question

When an electric current is passed through acidulated water, 112 ml. of hydrogen gas at STP collects at the cathode in 965 second. The current passed, in ampere is:

(A) 1.0

(B) 0.5

(C) 0.1

(D) 2.0

Question

Match List-I with List-II.

List-I (Conversion)	List-II (Number of Faraday required)
A. 1 mol of H_2O to O_2	I. 3 F
B. 1 mol of MnO_4^- to Mn^{2+} to Mn^{2+}	II. 2 F
C. 1.5 mol of Ca from molten $CaCl_2$	III. 1 F
D. 1 mol of FeO to Fe_2O_3	IV. 5 F

(A) A-II, B-III, C-I, D-IV

(B) A-III, B-IV, C-II, D-I

(C) A-II, B-IV, C-I, D-III

(D) A-III, B-IV, C-I, D-II

Question

The same amount of electricity was passed through two separate electrolytic cells containing solutions of nickel nitrate $[Ni(NO_3)_2]$ and chromium nitrate $[Cr(NO_3)_3]$ respectively. If 0.3 g of nickel was deposited in the first cell, the amount of chromium deposited is:
(at. wt. of Ni = 59, at. wt. of Cr = 52)

(A) 0.1 g

(B) 0.17 g

(C) 0.3 g

(D) 0.6 g

Question

A factory produces 40 kg of calcium in two hours by electrolysis. How much aluminium can be produced by the same current in two hours:
(At wt. of Ca = 40, Al = 27)

A) 22 kg

B) 18 kg

C) 9 kg

D) 27 kg

Question

What would be the ratio of moles each of Ag^+, Cu^{+2}, Fe^{+3} ions would be deposited by passage of same quantity of electricity through solutions of their salts:

(A) $1 : 1 : 1$

(B) $1 : \frac{1}{2} : \frac{1}{3}$

(C) $\frac{1}{3} : \frac{1}{2} : 1$

(D) $1 : 2 : 3$

Question

A solution of sodium sulphate in water is electrolysed using inert electrodes. The product at the cathode and anode are respectively:

(A) H_2, SO_2

(B) O_2, H_2

(C) O_2, Na

(D) H_2, O_2

Question

The passage of current liberates H_2 at cathode and Cl_2 at anode the solution is:

(A) $CuSO_4$(aq)

(B) $CuCl_2$(aq.)

(C) NaCl(aq.)

(D) Water